

Changes in Travel Behaviour to Education and Work in New Zealand

A Comparative Analysis of Census 2018 and 2023 Data



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DATA 480 - Research Methods

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Presentation Outline

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Research Goals

Introduction & Motivation

The “Journey to Education” Gap

- **Impact:** Shapes peak demand, congestion, and emissions.
- **Historical Bias:** Policy focuses heavily on “Journey to Work”.
- **Education Travel is:**
 - Substantial in scale
 - Behaviourally distinct
 - Under-analysed in national studies



The Post-COVID Context

Pandemic Disruption

The pandemic caused structural shifts in daily mobility:

- Widespread remote work & study
- Decline in public transport usage
- Potential long-term behavioural changes



Key Motivation

Do education and work travel show similar or different post-COVID recovery patterns?

Research Objectives

Research Questions

1

Primary Question

How have travel-mode distributions for education and work changed in New Zealand between the 2018 and 2023 Censuses, and do these changes differ systematically?

2

Secondary Focus

- Are changes uniform across space?
- Do urban and non-urban contexts show different patterns?
- Are observed differences statistically meaningful?

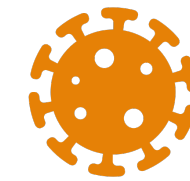
Literature Background

Key Themes & Identified Gap



Journey Purpose Matters

Education and work travel show systematically different mode-choice behaviours.



Uneven COVID Effects

Strong declines in public transport and persistence of remote participation.



Spatial Context Critical

Urban vs. non-urban differences strongly shape mode choice.



The Research Gap

Limited national-scale, population-level comparison of education vs work travel in NZ post-COVID.

Data Sources

Statistics New Zealand Census Data

High-resolution Census data from two key periods: 2018 & 2023

- Core Datasets:
 - Main Means of Travel to Education.
 - Main Means of Travel to Work.
- Why Census Data?
 - National coverage, consistent measurement,
 - suitable for distributional analysis.



Data Sources: NZ Census

15

Unique Travel Modes

10

Education

11

Work

1M+

Records
(Aggregated)

2

Census Years
(2018 & 2023)

2.4k

Statistical Areas (SA2)

68

Districts/City

17

Regions

Data Preparation & Harmonisation



Harmonise

Align education and work datasets into a common analytical structure.



Recode Categories

Group detailed modes into 5 policy groups: Private, Public, Active, At Home, Other.



Data Quality

Explicit handling of confidentiality suppression and structural zeros.



Ensure valid, unbiased comparison across journey types and years.

Proposed Methodology

Analytical Stages

1

Descriptive Analysis

Compute mode share & percentage-point changes (2018 vs 2023).

2

Distributional Analysis

Compare multinomial distributions using Chi-square tests of homogeneity.

3

Spatial Comparison

Identify differential recovery trajectories in Urban vs Non-urban contexts.

Research Goals & Visual Support

Core Goals

- Quantify travel-mode changes (2018-2023).
- Compare education vs work journeys.
- Examine spatial variation (Urban/Non-urban).
- Assess demographic heterogeneity.

Visual Output

Complementing inference with:



Comparative Bar Charts



Sankey Flow Diagrams



Regional Views

Expected Contribution



Academic Contribution

- National-scale comparative evidence.
- Education travel analysed explicitly, not as an afterthought.
- Rigorous distributional statistical framing.



Policy Relevance

- Inform transport planning beyond commuting.
- Identify distinct policy needs for education travel.
- Support post-COVID investment decisions.

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